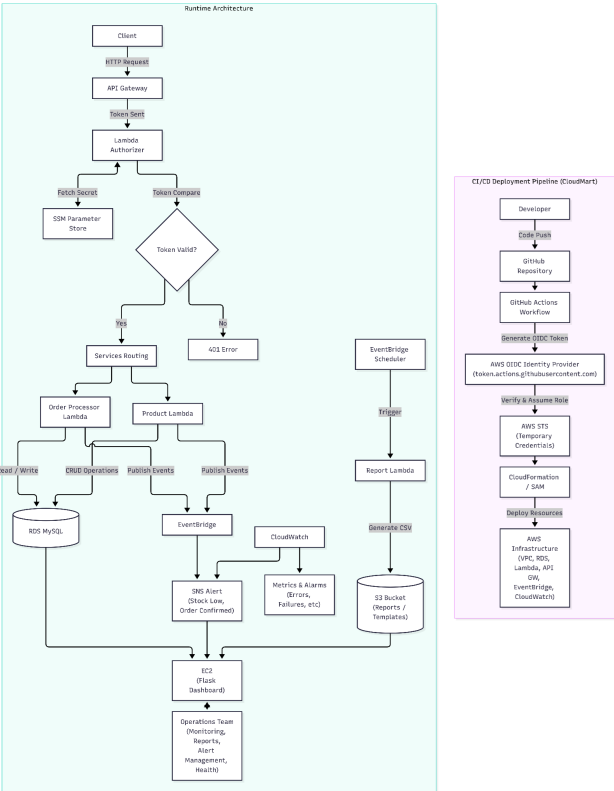


CloudMart Runtime & CI/CD Architecture

Concise architecture explanation based on the provided deployment and runtime diagram.



1. Architecture Overview

The architecture has two major parts: **Runtime Architecture** and the **CI/CD Deployment Pipeline**. Runtime handles API requests, authorization, database operations, reporting, monitoring, alerts, and the Flask dashboard. The CI/CD pipeline automates deployment of the AWS infrastructure.

2. Runtime Request Flow

Client → API Gateway → Lambda Authorizer → Services Routing → Order/Product Lambda → RDS MySQL

The client sends an HTTP request through API Gateway. The Lambda Authorizer receives the token, retrieves the required secret from SSM Parameter Store, and validates the token. Valid requests continue to service routing; invalid requests receive a **401 Error**.

3. Application Services

Component	Role
Order Processor Lambda	Processes order-related operations and reads/writes order data in RDS.
Product Lambda	Handles product operations and interacts with RDS.
RDS MySQL	Central application database for customers, products and orders.
EventBridge	Receives application events and triggers downstream processing.
SNS Alert	Sends alerts such as low-stock and order-confirmation notifications.

4. Reporting & Dashboard Flow

EventBridge Scheduler → Report Lambda → Generate CSV → S3 Bucket → EC2 Flask Dashboard

The scheduled Report Lambda generates CSV reports and stores them in S3. The EC2-hosted Flask dashboard uses the database and report storage to provide operational visibility to the Operations Team.

5. Monitoring and Alerts

CloudWatch collects metrics and supports alarms for errors and failures. Application events can flow through **EventBridge** to **SNS Alerts** for operational notifications such as low-stock conditions and order confirmations.

6. CI/CD Deployment Pipeline

Developer → GitHub Repository → GitHub Actions → AWS OIDC Identity Provider → AWS STS → CloudFormation / SAM → AWS Infrastructure

A code push starts the GitHub Actions workflow. The workflow generates an OIDC token, AWS verifies the identity and provides temporary STS credentials. CloudFormation/SAM then deploys the required AWS resources.

7. Security Flow

- **Token-based access:** API requests pass through the Lambda Authorizer.
- **Secret retrieval:** the authorizer fetches the required secret from SSM Parameter Store.
- **Authorization decision:** valid tokens proceed; invalid tokens return 401.
- **CI/CD authentication:** GitHub Actions uses OIDC and temporary AWS credentials rather than long-term AWS keys.

8. End-to-End Architecture

Users → API Gateway → Authorizer → Lambda Services → RDS
RDS / Events → EventBridge / SNS / CloudWatch
EventBridge Scheduler → Report Lambda → S3 → Flask Dashboard → Operations Team
Developer → GitHub → Actions + OIDC → STS → CloudFormation/SAM → AWS

CloudMart Runtime & CI/CD Architecture • Based on the provided architecture diagram